

# GPA-Aligned LoRA Merging via Generalized Procrustes Analysis

## Weeks 1–2 Progress Report

*Adapter Analysis & Initial Merging Comparison*

### Part 1: Week 1 Pre-Alignment Adapter Analysis

#### 1.1 Overview

This section reports the findings from the post-training inspection and norm analysis of the five QLoRA adapters trained on GLUE tasks (SST-2, MNLI, QNLI, CoLA, and RTE) using Qwen2.5-1.5B as the base model. The purpose of this analysis was to characterise the raw, pre-alignment structure of the adapter weight matrices before any merging or GPA alignment was applied.

The central question motivating this step was whether independently trained LoRA adapters differ substantially in scale and internal structure. If they do, this provides direct empirical motivation for the GPA-based alignment approach proposed in the dissertation, whose equal-contribution property should prevent larger-norm adapters from dominating the merged consensus.

The analysis examines adapter norms from multiple angles: task-level aggregates, module-family breakdowns (attention vs. MLP), per-layer heatmaps, depth profiles, and the relationship between adapter magnitude and standalone task performance. Five figures were generated, and each is discussed below.

#### 1.2 Per-Task Adapter Norm Ranking

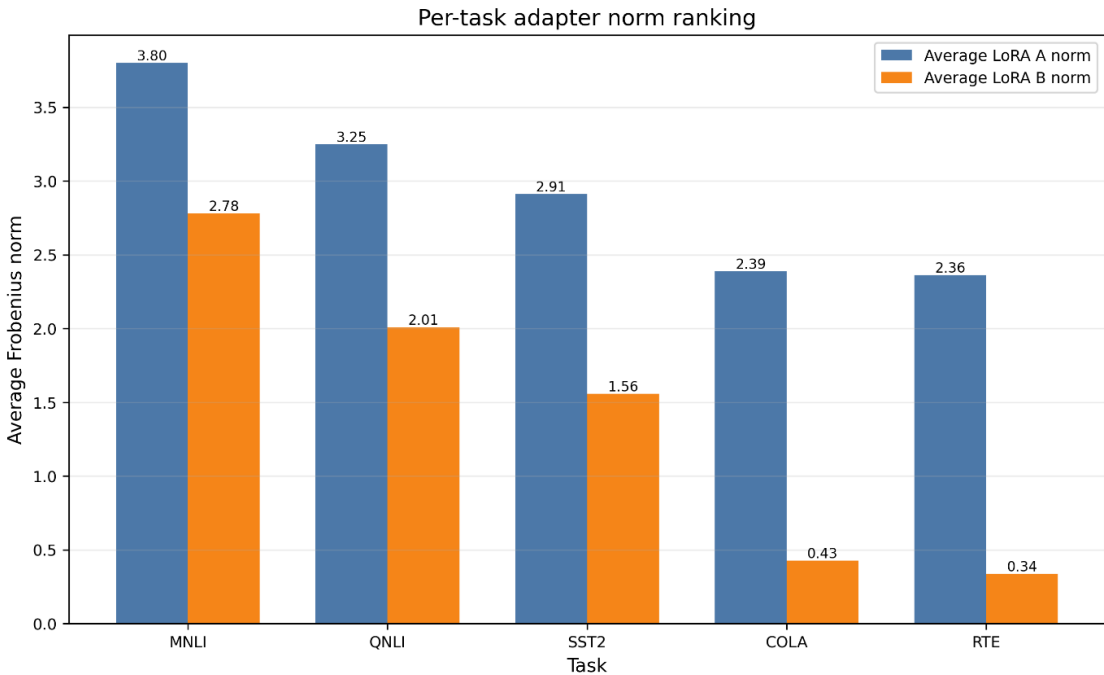


Figure 1: Per-task adapter norm ranking. Average Frobenius norms of LoRA A and LoRA B matrices across all modules, ordered by overall magnitude.

The norm ranking figure provides the most direct evidence of scale imbalance across adapters. MNLI has the largest average LoRA A norm at 3.80, followed by QNLI at 3.25 and SST-2 at 2.91. CoLA and RTE cluster at the bottom with norms of approximately 2.39 and 2.36 respectively. The LoRA B norms follow a similar ranking but with much greater disparity: MNLI’s B norm of 2.78 is over eight times larger than RTE’s 0.34.

This imbalance is significant for merging. In a naive Task Arithmetic merge (simple averaging of task vectors), the MNLI adapter would contribute disproportionately to the merged result purely by virtue of its larger parameter magnitudes, not because its learned features are inherently more important. The roughly 8× spread in LoRA B norms between MNLI and RTE means that, without any normalisation or alignment, the RTE adapter’s contributions would be almost entirely overwhelmed.

This finding directly supports the motivation for GPA-based merging: GPA’s equal-contribution property ensures that each adapter’s rotated factor matrix contributes equally to the consensus, regardless of its raw Frobenius norm. The norm ranking figure establishes the empirical baseline that this property is needed.

### 1.3 Attention vs. MLP Module Family Breakdown

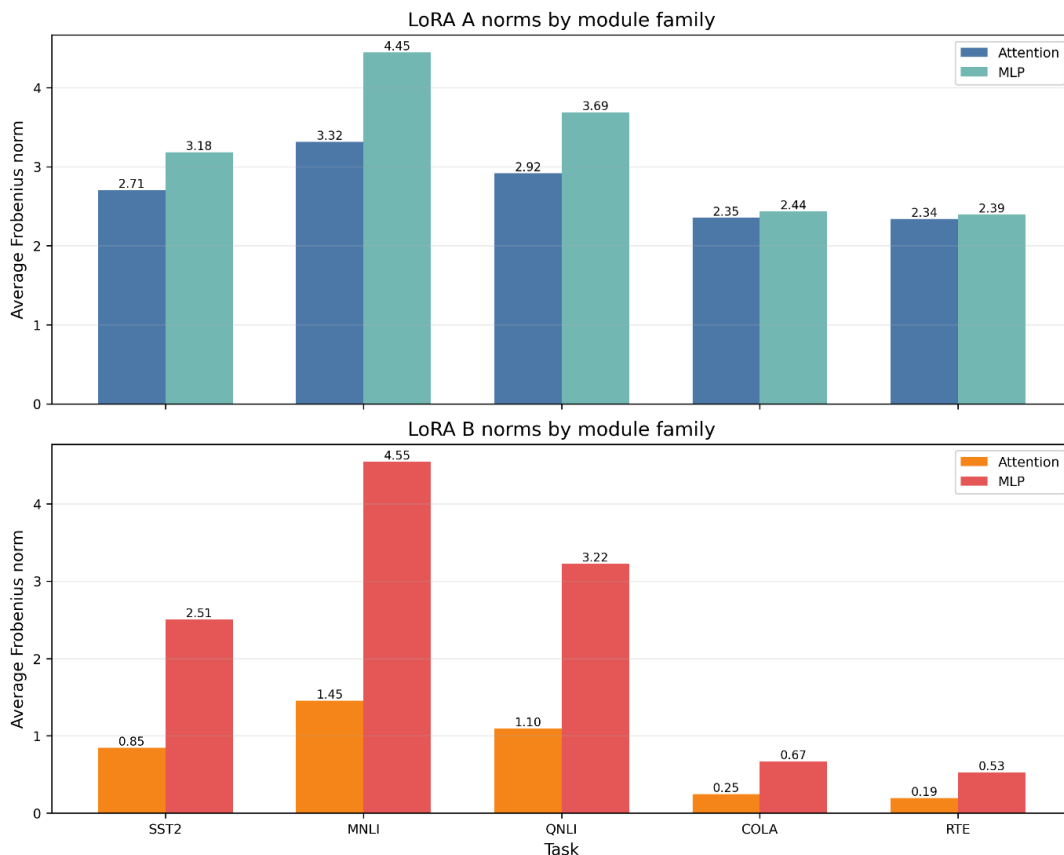


Figure 2: LoRA A and LoRA B norms decomposed by module family (attention vs. MLP) for each task.

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The module-family breakdown reveals that the norm imbalance is not simply a global scaling effect applied uniformly across all parameters. For LoRA A, MLP modules consistently carry larger norms than attention modules across all five tasks. The gap is particularly pronounced for MNLI (attention: 3.32, MLP: 4.45) and QNLI (attention: 2.92, MLP: 3.69), while CoLA and RTE show a much smaller attention–MLP differential.

The LoRA B norms tell a complementary story. The MLP-to-attention ratio is even more extreme: MNLI’s MLP LoRA B norm (4.55) is over three times its attention B norm (1.45). This suggests that the MLP layers undergo more substantial adaptation than the attention layers, and that this effect is task-dependent such that larger-data tasks (MNLI, QNLI) adapt their MLP pathways more aggressively.

For the merging pipeline, this structured imbalance means that a single per-adapter scaling factor would be insufficient to balance contributions. The GPA alignment, which operates independently on each LoRA module, is well-suited to handle this heterogeneity because it computes separate rotations for each of the 196 LoRA modules (28 layers  $\times$  7 module types) across the network.

## 1.4 Per-Layer Norm Heatmap

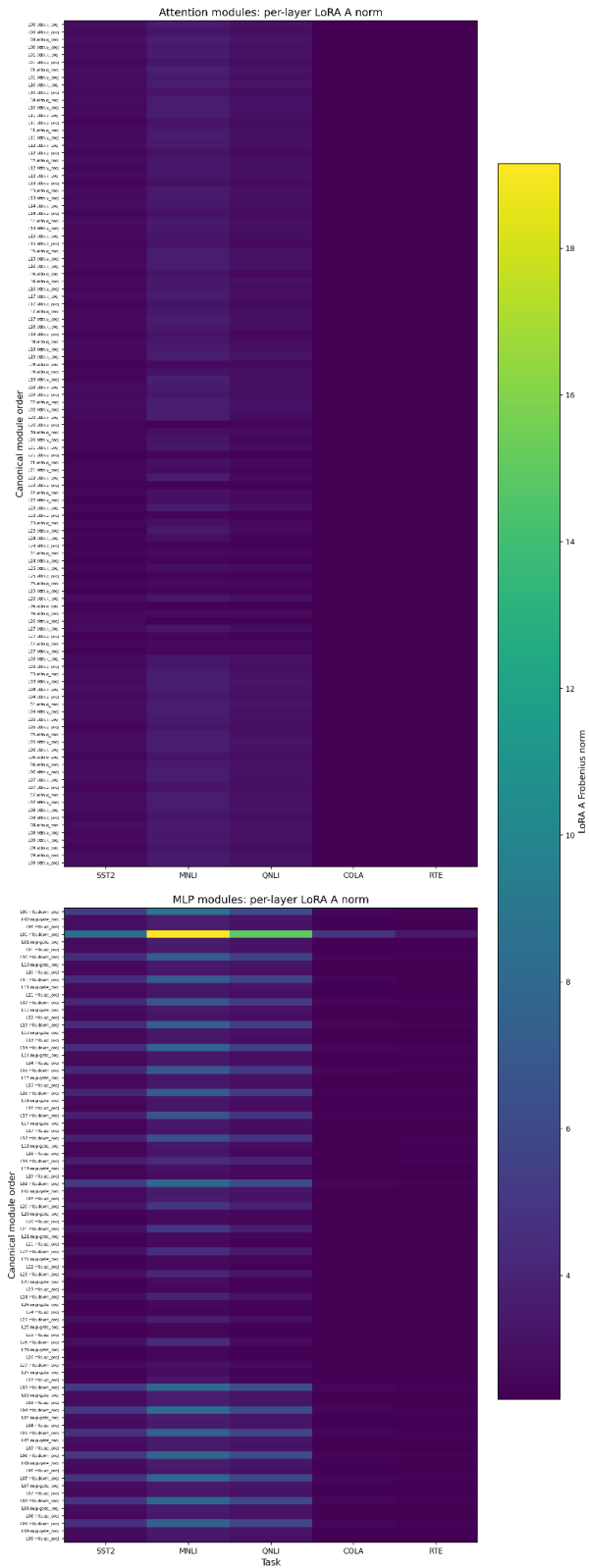


Figure 3: Per-layer LoRA A Frobenius norm heatmap for attention modules (top) and MLP modules (bottom), across all five tasks.

The layer-level heatmap is the strongest descriptive figure for demonstrating that adapter structure is heterogeneous and locally varying. MNLI and QNLI show consistently elevated norms across most layers, but with visible variation because certain layers appear as brighter bands, particularly in the MLP panel where early layers (approximately layers 0–4)

show notably high norms. CoLA and RTE appear as uniformly dark columns, confirming that these adapters contribute much smaller parameter updates across the entire network depth.

The MLP panel shows more pronounced layerwise variation than the attention panel, with distinct horizontal banding patterns that differ between MNLI and QNLI, suggesting that these tasks concentrate their MLP adaptation at different layer depths. This layer-local heterogeneity provides the strongest argument that merging methods must account for structural differences at a fine granularity. A global rescaling or single alignment transformation per adapter would miss the layer-specific patterns visible in this heatmap.

## 1.5 Depth Profiles

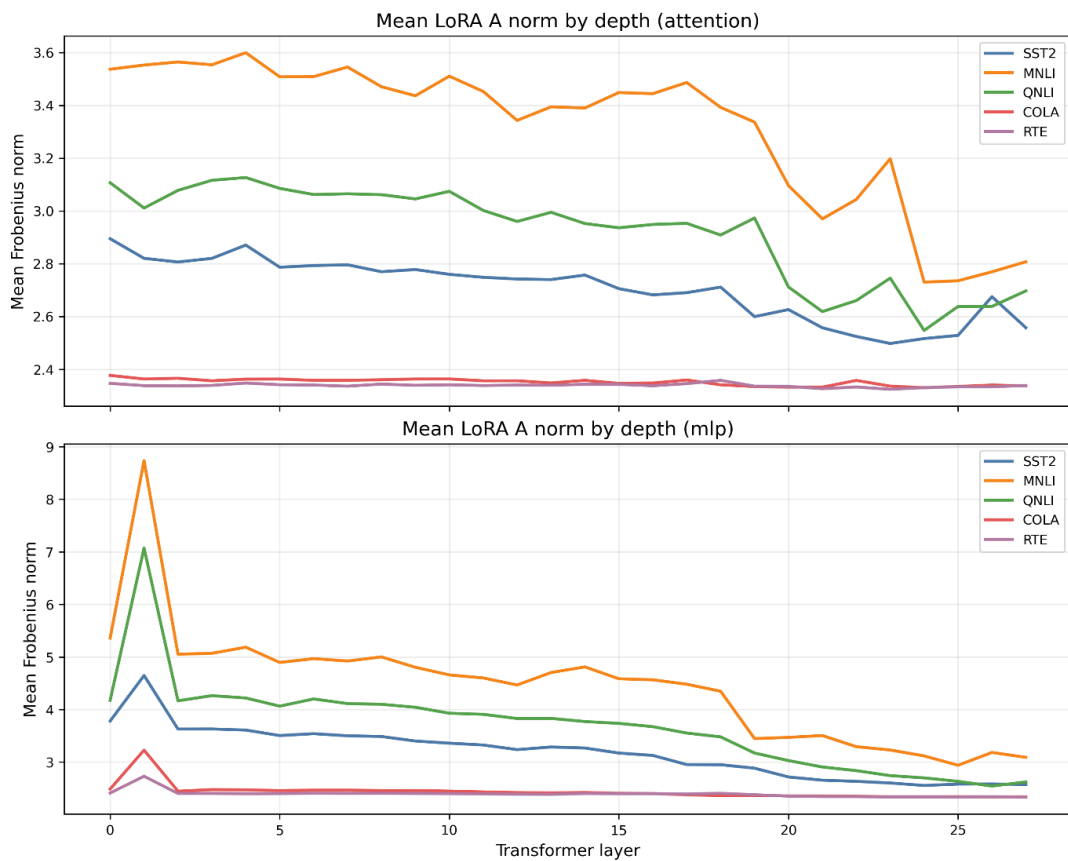


Figure 4: Mean LoRA A Frobenius norm versus transformer layer depth, for attention modules (top) and MLP modules (bottom).

The depth-trend figure complements the heatmap by showing continuous profiles. In the attention panel, MNLI maintains the highest norm throughout the network but shows a gentle downward trend from approximately 3.55 at layer 0 to 2.80 at layer 28. QNLI follows a similar declining pattern but at lower absolute values. SST-2 is relatively flat. CoLA and RTE are nearly indistinguishable, both flat at approximately 2.35–2.40.

The MLP panel reveals a more dramatic pattern. MNLI shows a sharp spike at layers 1–2 (norm approaching 9.0), followed by a steady decline to approximately 3.0 by layer 28. QNLI mirrors this early-layer spike at a smaller scale (peaking near 7.0). These depth profiles suggest that larger-data tasks concentrate heavy adaptation in the early MLP layers, potentially modifying the model’s input processing and feature extraction pathways. The later layers show more uniform, lower-magnitude adaptation.

## 1.6 Standalone Performance vs. Adapter Norm

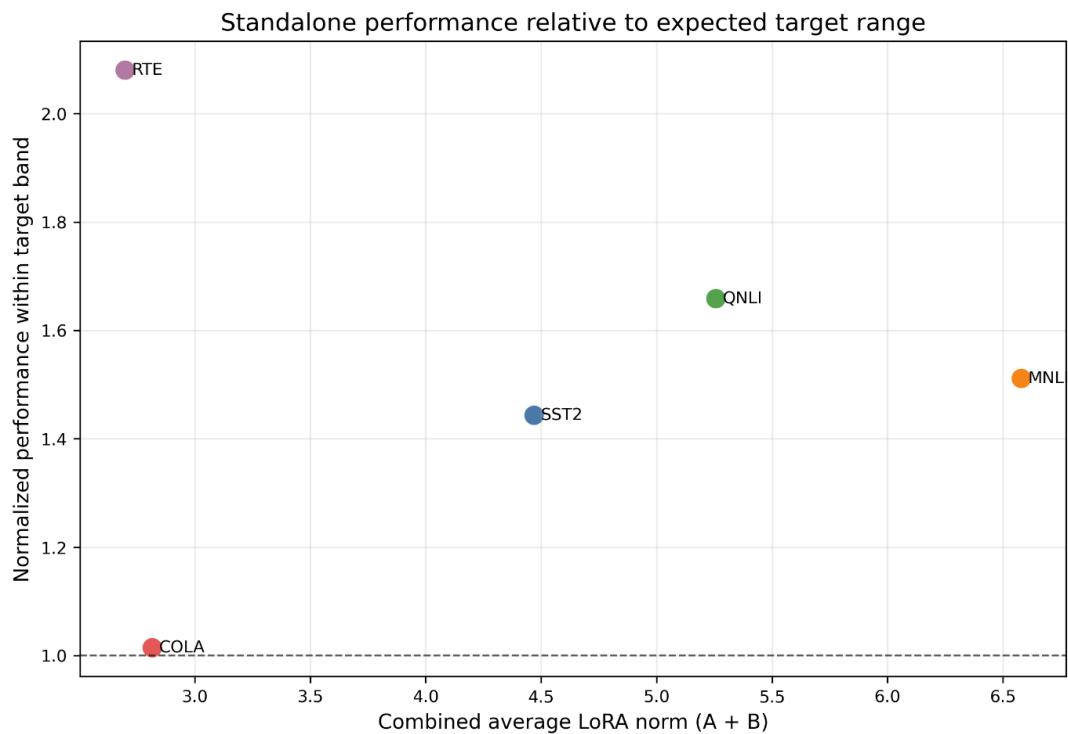


Figure 5: Combined average LoRA norm (A + B) versus normalised standalone validation performance for each task.

The scatter plot of performance versus norm is treated as an exploratory, secondary figure. The relationship between adapter magnitude and standalone performance does not show a clear linear trend. RTE, despite having the smallest combined norm (approximately 2.8), achieves the highest normalised performance (approximately 2.1), while MNLI, with the largest norm (approximately 6.5), sits at a normalised performance of roughly 1.5.

The absence of a strong positive correlation between norm and performance supports an important point: adapter magnitude alone is not a reliable proxy for adapter quality or the importance of the task’s learned features. This reinforces the argument that naive magnitude-based weighting in merging is inappropriate, and that alignment-based approaches such as GPA, which focus on geometric structure rather than raw scale, are better motivated.

## 1.7 Summary of Numerical Findings

| Metric          | MNLI | QNLI | SST-2 | CoLA | RTE  |
|-----------------|------|------|-------|------|------|
| Avg LoRA A norm | 3.80 | 3.25 | 2.91  | 2.39 | 2.36 |
| Avg LoRA B norm | 2.78 | 2.01 | 1.56  | 0.43 | 0.34 |
| LoRA A (Attn)   | 3.32 | 2.92 | 2.71  | 2.35 | 2.34 |
| LoRA A (MLP)    | 4.45 | 3.69 | 3.18  | 2.44 | 2.39 |
| LoRA B (Attn)   | 1.45 | 1.10 | 0.85  | 0.25 | 0.19 |
| LoRA B (MLP)    | 4.55 | 3.22 | 2.51  | 0.67 | 0.53 |

|                |      |      |      |      |      |
|----------------|------|------|------|------|------|
| Combined (A+B) | 6.58 | 5.26 | 4.47 | 2.82 | 2.70 |
|----------------|------|------|------|------|------|

## 1.8 Implications for GPA-Based Merging

Taken together, these five figures established a clear empirical case for the necessity of alignment-aware merging. The key findings were:

1. **Scale imbalance is substantial.** The combined norm spread between the largest (MNLI, 6.58) and smallest (RTE, 2.70) adapters is approximately 2.4×. For LoRA B norms alone the ratio exceeds 8×. Naive averaging would heavily bias the merged model towards MNLI’s learned features.
2. **The imbalance is structured by module family.** MLP modules consistently have larger norms than attention modules, and this gap is itself task-dependent. A single per-adapter rescaling cannot capture this structure.
3. **The imbalance is layer-local and heterogeneous.** Early MLP layers show dramatically higher norms for the larger-data tasks, while deeper layers converge. Alignment pressure will vary across the network depth.
4. **Norm is not a proxy for quality.** The weak relationship between adapter magnitude and standalone performance means that magnitude-based weighting in merging is not well-justified. Geometric alignment via GPA focuses on structural similarity rather than raw scale.

These observations provided the empirical foundation for Week 2, where GPA alignment would be applied to these real adapters and the merging baselines would be implemented and compared against GPA+TIES factor-space merging.

## 1.9 Expectations for Week 2

At the close of Week 1, the immediate next steps were:

5. Implement the baseline merging methods: Task Arithmetic, TIES-Merging, DARE, and LR-KnOTS.
6. Apply GPA alignment to the real LoRA A matrices across all 196 modules and record per-layer diagnostics.
7. Run the first GPA+TIES factor-space merge and evaluate on all five GLUE tasks.
8. Begin writing the Introduction and Related Work sections of the dissertation.

The norm imbalances documented above were expected to serve as the primary descriptive evidence in the dissertation’s motivating section, and the per-layer diagnostics from GPA alignment would be compared against these baseline profiles to assess whether alignment successfully reduces the structural disparities.

## Part 2: Week 2 — Initial Merging Comparison

### 2.1 Summary

All four baseline merging methods and the proposed GPA+TIES pipeline were successfully implemented and compared at a fixed hyperparameter configuration during Week 2, fulfilling the first three objectives set at the end of Week 1. The Introduction and Related Work draft was also begun, addressing the fourth objective. This section reports the results of the initial comparison and interprets them in light of the adapter analysis from Week 1.

The headline finding is that **GPA + TIES is the strongest factor-space TIES-style method in this first comparison**, outperforming both unaligned TIES-Merging and the LR-KnOTS baseline. However, the simple Task Arithmetic baseline still achieves the highest overall average, and the GPA advantage is concentrated in a single task rather than being broadly consistent. These are early results at a single, likely suboptimal, hyperparameter setting, and the Week 3 sweeps should give a much clearer picture.

### 2.2 Experimental Setup

All methods were compared at a single fixed configuration to establish a first data point before the full hyperparameter sweep: scaling coefficient  $\lambda = 1.0$ , trim percentage 20%, DARE drop probability 0.1, and total majority sign election.

The average primary score is the arithmetic mean of each task’s primary GLUE metric: accuracy for SST-2, MNLI, QNLI, and RTE, and Matthews correlation for CoLA. All methods merge five rank-16 QLoRA adapters trained on Qwen2.5-1.5B.

### 2.3 Results

#### Overall Ranking

| Method            | Avg Primary Score |
|-------------------|-------------------|
| Task Arithmetic   | 0.3900            |
| <b>GPA + TIES</b> | <b>0.3750</b>     |
| DARE + TIES       | 0.3746            |
| TIES-Merging      | 0.3619            |
| LR-KnOTS + TIES   | 0.3607            |

GPA + TIES scores 0.3750, placing it above TIES-Merging (+0.0131), LR-KnOTS + TIES (+0.0143), and marginally above DARE + TIES (+0.0004). Task Arithmetic leads at 0.3900, though it produces a rank-80 merged adapter rather than a compact rank-16 one.

#### Per-Task Breakdown

| Method               | SST-2  | MNLI   | QNLI   | CoLA   | RTE    | Avg    |
|----------------------|--------|--------|--------|--------|--------|--------|
| Oracle (single-task) | 0.9633 | 0.8955 | 0.9458 | 0.6522 | 0.8664 | 0.8646 |



|                   |               |               |               |               |               |               |
|-------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Task Arithmetic   | 0.4897        | 0.4350        | 0.4946        | 0.0000        | 0.5307        | 0.3900        |
| TIES-Merging      | 0.4312        | 0.3567        | 0.4944        | 0.0000        | 0.5271        | 0.3619        |
| DARE + TIES       | 0.4908        | 0.3609        | 0.4944        | 0.0000        | 0.5271        | 0.3746        |
| LR-KnOTS + TIES   | 0.4472        | 0.3346        | 0.4944        | 0.0000        | 0.5271        | 0.3607        |
| <b>GPA + TIES</b> | <b>0.5092</b> | <b>0.3443</b> | <b>0.4944</b> | <b>0.0000</b> | <b>0.5271</b> | <b>0.3750</b> |

## 2.4 Key Findings

### 2.4.1 GPA + TIES is the best factor-space method, but the gain is narrow

Relative to unaligned TIES-Merging, GPA + TIES improves SST-2 substantially (0.5092 vs 0.4312, a gain of +0.078), but is slightly worse on MNLI (0.3443 vs 0.3567, a drop of -0.012) and effectively unchanged on the other three tasks. The proposal's success criterion requires improvement consistent across at least 3 of 5 tasks, which is not yet met.

*Reflecting on the Week 1 analysis:* the adapter norm analysis predicted that MNLI would dominate under naive merging due to its substantially larger norms (combined A+B of 6.58 versus RTE's 2.70). The fact that GPA + TIES improves SST-2—a mid-range adapter in terms of norm (4.47)—while slightly degrading MNLI is consistent with the hypothesis that alignment redistributes influence away from the dominant adapter. However, the narrow scope of the improvement (only one task) suggests that at  $\lambda = 1.0$  the alignment effect may be overshadowed by the overscaling problem discussed below.

### 2.4.2 CoLA collapses completely for every merged method

Every merged adapter achieves a Matthews correlation of exactly 0.0 on CoLA, despite the single-task adapter scoring 0.6522. The merged models appear to produce near-constant predictions on this task. This is believed to be primarily a  $\lambda$ -scaling problem: at  $\lambda = 1.0$ , the merged adapter may be so far from the pretrained weights that CoLA predictions degenerate.

*Reflecting on the Week 1 analysis:* CoLA had one of the smallest adapter norms (combined A+B of 2.82) and showed the least layerwise variation in the heatmap analysis. It is perhaps unsurprising that CoLA is the most vulnerable to interference from the much larger MNLI and QNLI adapters during merging. The complete collapse to 0.0 Matthews correlation for all methods suggests a systematic issue with the scaling coefficient rather than a method-specific failure. The Week 1 finding that norm is not a proxy for quality is particularly relevant here: CoLA's small norm does not mean its learned features are unimportant, but it does mean they are easily overwhelmed.

### 2.4.3 QNLI and RTE do not discriminate between methods

QNLI ( $\approx 0.494$ ) and RTE ( $\approx 0.527$ ) are nearly identical across all methods, suggesting that  $\lambda = 1.0$  pushes every method into a similarly poor prediction regime on these tasks. A lower  $\lambda$  should expose meaningful method differences.

### 2.4.4 Factor-space methods are overscaled at $\lambda = 1.0$

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Evaluation losses tell the clearest story. Task Arithmetic’s MNLI loss is 1.13, while GPA + TIES is at 2.19 and LR-KnOTS + TIES at 2.74. This strongly suggests the factor-space methods need a lower  $\lambda$  to perform at their best. The Week 3 sweep will use a finer grid in the 0.05–0.5 range for these methods.

*Reflecting on the Week 1 analysis:* the depth profile analysis showed that early MLP layers in MNLI and QNLI adapters had dramatically elevated norms (MNLI peaking near 9.0 at layers 1–2). When these large-magnitude modules are merged at full scale ( $\lambda = 1.0$ ), the resulting perturbation to the pretrained model is likely too severe. The per-module nature of GPA alignment should help redistribute this energy more evenly, but only if the overall scaling is first brought into a reasonable range.

#### 2.4.5 TIES trimming does not produce sparse merged adapters

Despite trimming 20% of each source adapter’s parameters, the final merged adapters have nonzero fractions of approximately 0.999. The disjoint merge step repopulates positions across tasks. This means DARE + TIES’s benefit likely comes from regularisation during sparsification rather than from producing a compact sparse adapter. This is a methodological observation worth discussing in the dissertation.

#### 2.4.6 Task Arithmetic is strong but not a like-for-like comparison

Task Arithmetic achieves the best average (0.3900) and leads on MNLI, QNLI, and RTE. However, it produces a rank-80 merged adapter ( $5 \times \text{rank-16}$ ), whereas the factor-space methods maintain a standard rank-16 LoRA shape. The fairest comparison is between GPA + TIES and the other compact factor-space baselines, where GPA + TIES does lead.

*Reflecting on the Week 1 analysis:* the module-family breakdown showed that MLP norms were consistently larger than attention norms, and this ratio was itself task-dependent. Task Arithmetic’s advantage here may partly stem from the fact that it preserves each adapter’s full rank-16 contribution in the merged result (yielding rank-80), thereby avoiding the information compression that factor-space methods require. Whether GPA’s alignment can close this gap at an optimised  $\lambda$  is a key question for Week 3.

### 2.5 Overall Interpretation

These results provide **early but limited support** for the alignment hypothesis. GPA + TIES is promising relative to the other factor-space methods, but the current evidence does not yet support a strong claim of broad superiority. The most important takeaway is that  $\lambda = 1.0$  is clearly a poor operating point for the factor-space methods, and the Week 3 hyperparameter sweep is the critical next step.

If GPA + TIES performs well at an optimised  $\lambda$  and shows consistent improvement across multiple tasks, the dissertation narrative strengthens considerably. If the advantage remains concentrated in SST-2, the contribution will be framed more carefully around the conditions under which alignment helps.

### 2.6 Revisiting Week 1 Expectations

The four objectives set at the end of Week 1 were all met:

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- All five baseline merging methods were implemented and evaluated.
  - GPA alignment was applied to the real LoRA A matrices across all 196 modules.
  - The first GPA+TIES factor-space merge was run and evaluated on all five GLUE tasks.
  - Writing on the Introduction and Related Work sections was begun.

However, the empirical picture is more nuanced than what was anticipated at the end of Week 1. The adapter analysis correctly identified the norm imbalances that would challenge naive merging, and GPA + TIES does indeed lead among the compact factor-space methods. But the universal CoLA collapse and the narrow task-specificity of the GPA advantage (limited to SST-2 at this configuration) were not explicitly anticipated. The Week 1 analysis focused on structural motivation for alignment but did not predict the sensitivity of all methods to the scaling coefficient. The Week 3 hyperparameter sweep is therefore the most important immediate priority.

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## Immediate Next Steps (Week 3)

9. **Complete Introduction & Related Work draft** (targeting approximately 3,000 words).
10.  **$\lambda$  sweep for all methods.** Using a finer grid at the low end for factor-space methods:  $\lambda \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.7, 1.0\}$ .
11. **CoLA prediction distribution diagnostic.** Log class distributions at each  $\lambda$  to determine whether CoLA recovers or is structurally broken.
12. **Full results table and Methods chapter** (including a discussion of the TIES sparsity observation).

The project remains on track with the revised implementation plan. Week 3 begins tomorrow and the  $\lambda$  sweep is the highest priority.